GAME SENSE

ESPORTS PERFORMANCE & GAMING HEALTH ANALYTICS | PROBLEM STATEMENT

Project Overview

Game Sense – Esports Performance & Gaming Health Analytics is a data analytics project focused on exploring relationships between esports players' match performance, biometric indicators, and lifestyle habits. The dataset includes player-session records containing measures such as Actions Per Minute (APM), reaction time, match outcome, sleep duration, energy drink consumption, heart rate, Heart Rate Variability (HRV), screen brightness, and blue-light filter use.

Problem Statement

Esports datasets contain detailed information about players' gaming performance, sleep, playtime, energy drink use, and physiological signals. However, raw records do not by themselves make it easy to identify which factors are associated with winning or losing matches. It can also be difficult to compare these relationships across game genres or to explore whether performance differences align more closely with gameplay indicators—such as APM and reaction time—or recovery-related measures, such as sleep, heart rate, and HRV.

This project addresses the need to clean, analyze, and visualize esports session data so that patterns in player performance and gaming-related health indicators can be explored more clearly. The results are intended to support informed discussion and further investigation by players and coaches.

Key Questions

- Which sleep-duration groups show different observed win rates?
- How do APM and reaction time relate to match outcomes?
- How is energy drink consumption associated with average heart rate?
- How do win rates and HRV scores vary across game genres?
- How can an interactive dashboard make these patterns easier to explore?

Project Objective

Use Python-based data cleaning and exploratory data analysis, followed by Power BI visualization, to examine relationships between gaming performance, lifestyle habits, and biometric indicators across player-session records.

Scope Note

This is a descriptive and exploratory analytics project. The observed relationships are associations in the dataset and should not be interpreted as proof that a particular habit or biometric measure causes a match outcome.